

level 2

no. of sessions: 6

cleaning bot

The cleaning bot moves around and helps clean the floor automatically, just like a robot vacuum cleaner at home.



simple machine: wheel & axle
electronic parts: polarity reversal





Motor Driver

The **brain of a robot**. I listen to input devices like sensors and tell output devices like motors/LEDs when to turn ON/OFF.



IR Sensor

I am like a **robot's eyes**. I help robots see objects or follow a path without touching anything.



LDR Sensor

I am like a **light detector** who tells a robot if it is bright or dark and i am used in automatic lights.



Motor

I **change electricity into movement** and help wheels, fans, and robots spin and move.



DPDT Switch Controller

Like a **direction button** - i make a motor spin forward or backward without using coding.



Servo Motor

I am a **smart motor** that can stop at the exact position you want. I am used in robot arms, gates, and moving parts.



Servo Controller

I am like a **control room** for the servo motor and lets you move the servo easily to test how it works.



Battery & Battery Holder

I **give power** to the project, and my battery holder keeps the battery safe and connected. Together, we help the model work.



Jumper Wires

We are the **roads for electricity** who connect the different parts of the circuit, so power can travel.



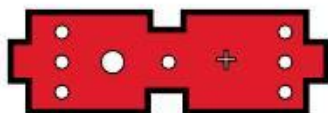
Motor Clamp

I am like a **strong hand** that holds the motor tightly in place while it spins.

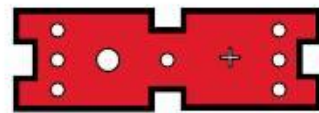
materials needed



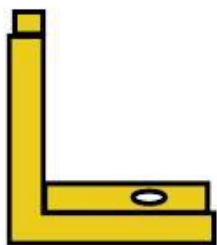
M-F beam - x2



M-M beam - x2



F-F beam - x1



L clamp - x2



I clamp - x2

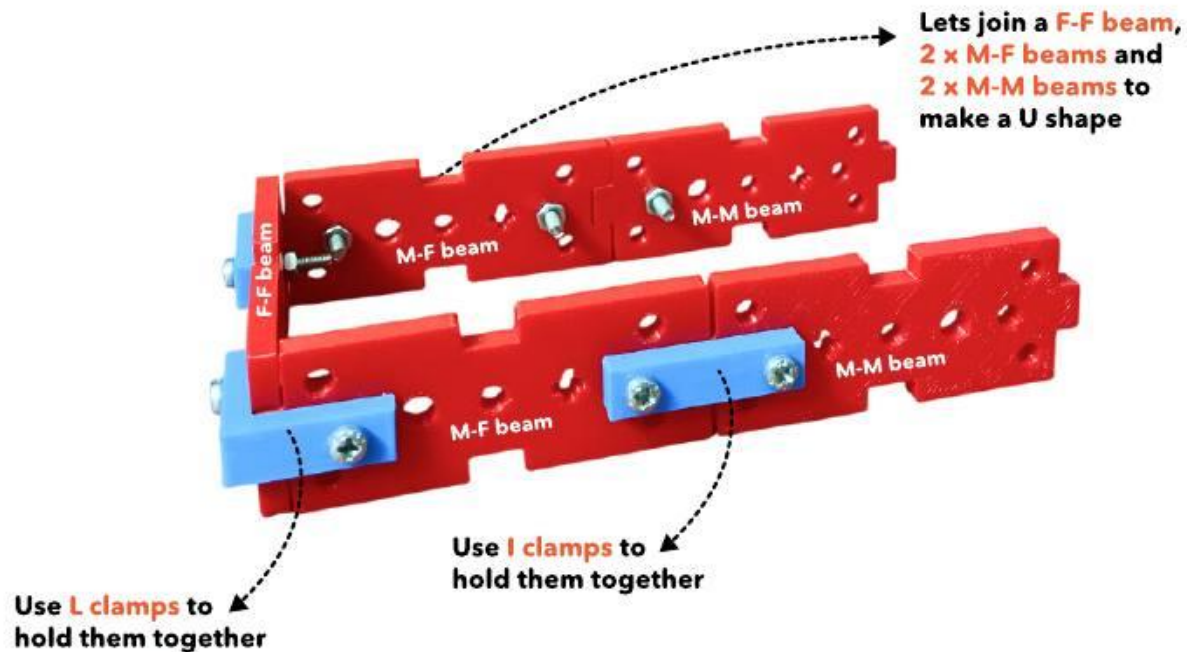


small screw - x8

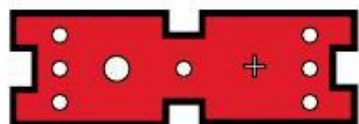


nut - x8

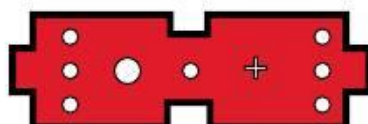
step 1 : assemble the base



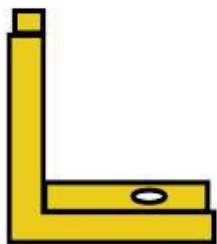
materials needed



F-F beam - x2



M-M beam - x1



L clamp - x2



I clamp - x2

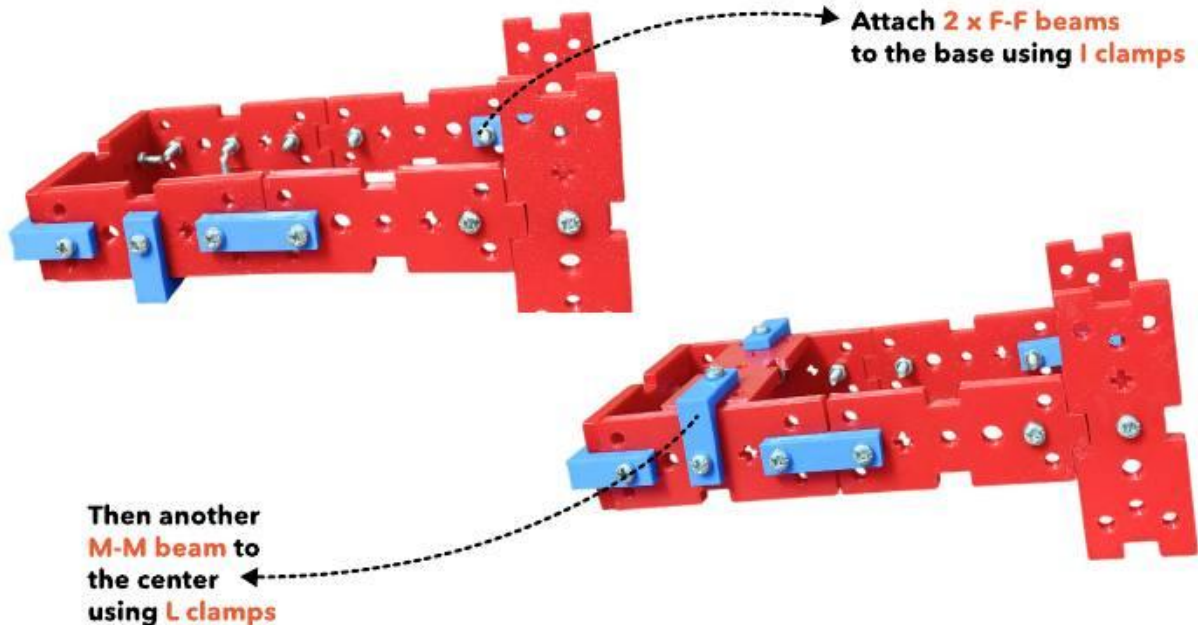


small screw - x8

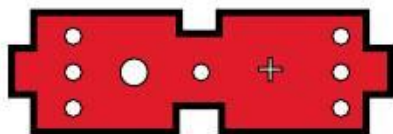


nut - x8

step 2: attach the supports



materials needed



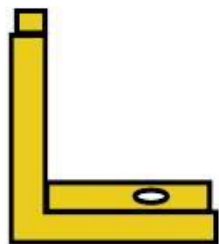
M-M beam - x1



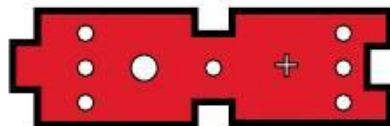
small screw - x14



nut -

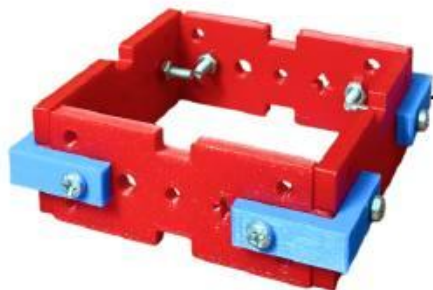


L clamp - x8



M-F beam - x4

step 3: assemble the motor holder



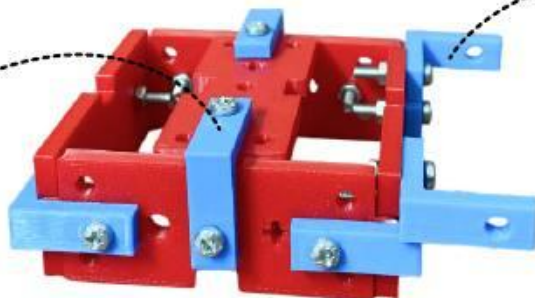
1.

Join 4 X M-F beams
to form a square.

Use L clamps
to hold them together.

2.

Attach another
M-M beam in the
middle using
L clamps as shown.



Add 2x L clamps
to the front of
the square shape
as shown.

materials needed



motor - x1

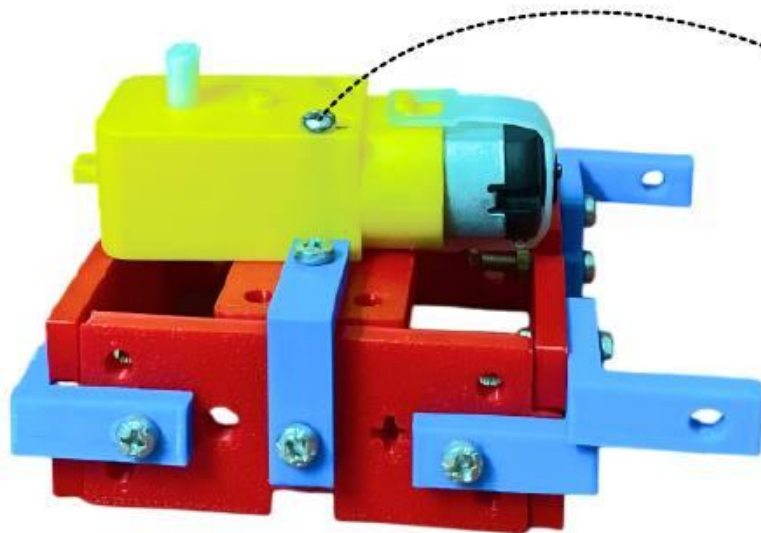


long screw - x1



nut - x1

step 4: attach the motor



Fix the motor to
the center of the
beam using one
long screw

materials needed

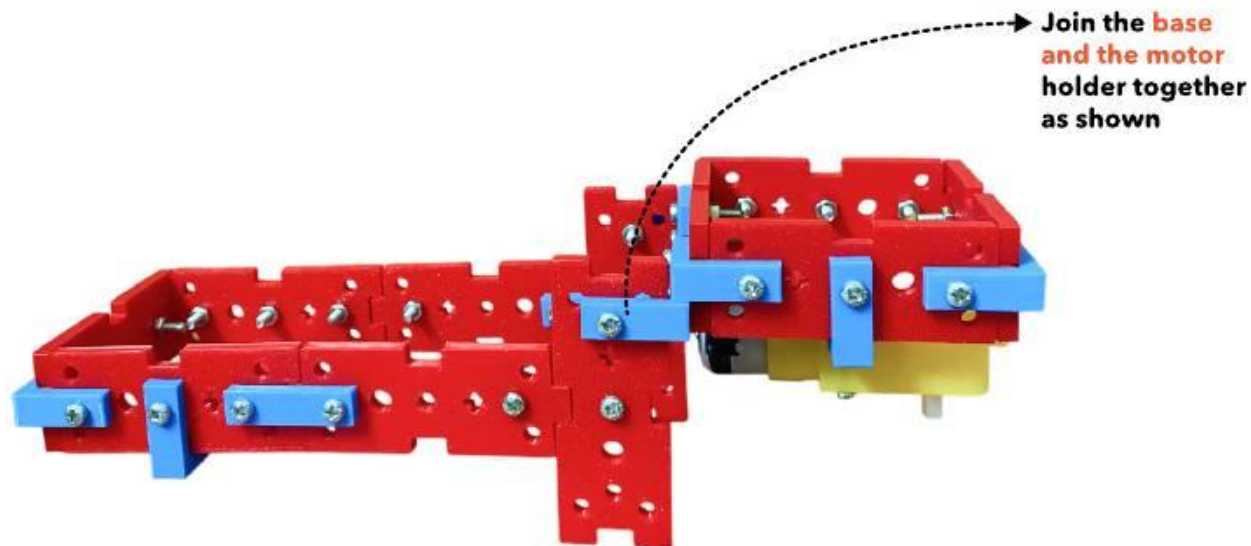


small screw - x2



nut - x2

step 5: attach motor holder to the base



materials needed



motor clamp - x2

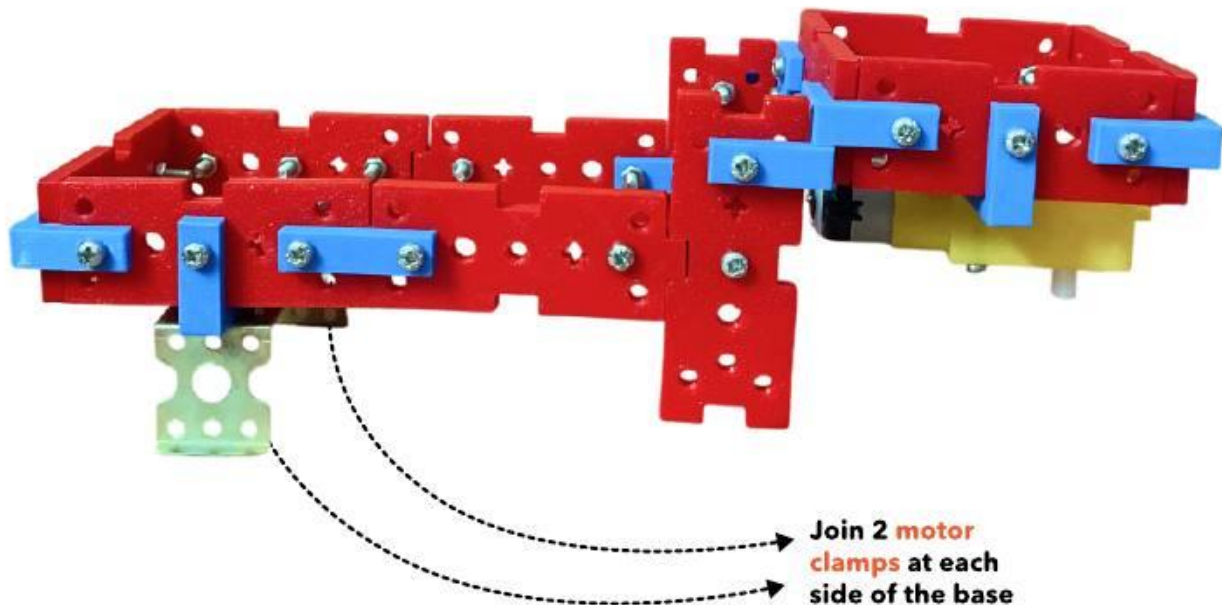


small screw - x4



nut - x4

step 6: add the motor clamps



materials needed



motor - x2

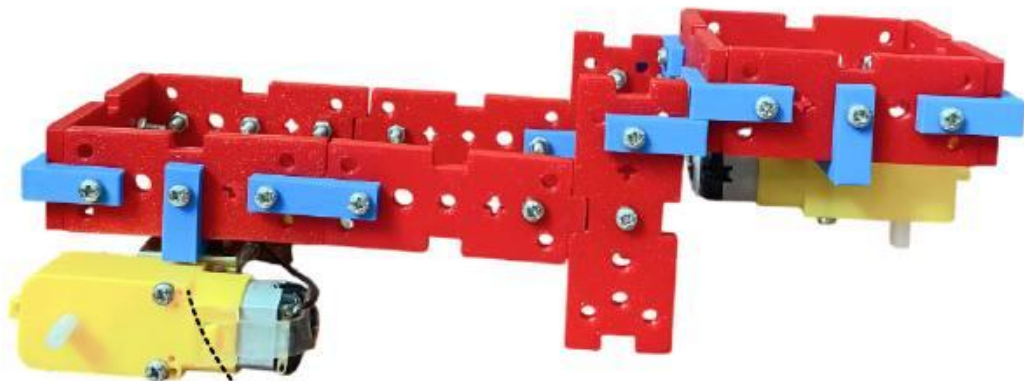


long screw - x2



nut - x2

step 7: assemble the back motors



Fix the motors to the clamps using long screw as shown.

materials needed



wheel - x4



long screw - x2



nut - x2

step 8: add the wheels



Add 2 wheels to the **beam in front**, using long screws.

Attach 2 wheels to the **back motor's axle**

materials needed



coupler - x1



I clamp - x4



long screw - x4



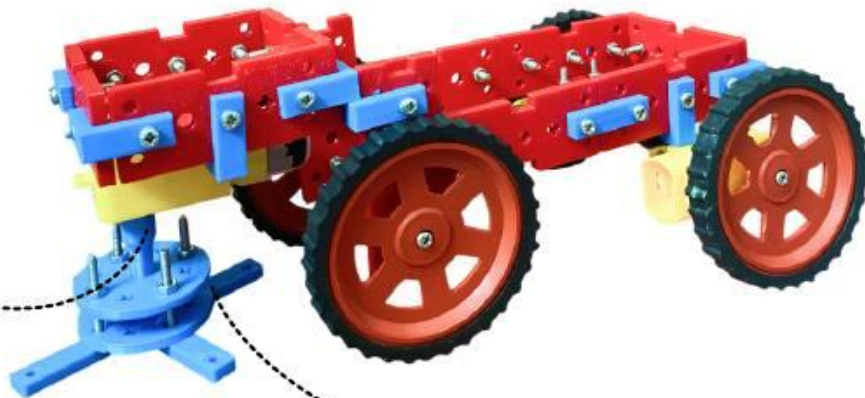
nut - x4

step 9: assemble the cleaning blades



Attach **4 x 1 clamps** to the coupler using long screws

Attach the coupler to the axle of the front motor



this cleaning blades help the bot push the dust away

materials needed



motor controller - x1



battery holder x1



battery x1

step 10: circuit

